

TALLY2EXCEL AUTOMATION

PROJECT DOCUMENTATION

Tally Sales Register → Formatted Excel Converter

Version 1.0 | June 2025

1. Executive Summary

Project Name	Tally2Excel Automation
Objective	To automate the conversion of raw Tally Sales Register XLSX exports into a clean, professionally formatted Excel file — without any server infrastructure, installation, or manual reformatting.
Type	Client-Side Browser Tool (Zero Backend)
Technology	HTML + CSS + JavaScript (SheetJS + ExcelJS)
Status	Version 1.0 — Production Ready

Objectives

- Eliminate manual copy-paste reformatting of Tally exports
- Auto-detect invoice columns and currency from Tally's native format
- Generate a styled, print-ready Excel file in seconds
- Ensure zero data leaves the user's device (100% local processing)
- Require no installation, server, or technical knowledge to operate

2. What is Tally2Excel Automation?

Tally2Excel Automation is a browser-based conversion tool that reads a raw Tally Sales Register XLSX export and produces a clean, structured Excel file with professional formatting — entirely within the user's browser, with no data upload to any server.

The tool automatically:

- Accepts a Tally-exported XLSX file via drag-and-drop or file picker
- Skips Tally's 11-row report header and reads actual invoice data
- Extracts Invoice Number, Buyer Name, Date, and Amount from fixed Tally columns
- Detects the currency symbol from the cell number format (INR, USD, EUR, GBP)
- Sorts records by Invoice Number in ascending order
- Generates a styled XLSX with purple headers, alternating row colors, borders, frozen header row, and auto-filter
- Displays a 5-row preview before download

Example:

Without Tally2Excel	With Tally2Excel
Accountant manually copies Tally export row-by-row into a formatted register, reformats dates, adds currency symbols, applies borders — taking 30–60 minutes per report.	User drops the Tally XLSX onto the tool. In under 5 seconds, a fully formatted Excel file is ready to download — no manual work required.

3. Business Problem Statement

Existing Situation

Businesses using Tally ERP for accounting regularly export the Sales Register for reconciliation, client sharing, audits, and MIS reports. The raw Tally export contains:

- 11 rows of report metadata (company name, date range, filters) before data begins
- Unformatted date serial numbers instead of readable dates
- No visual formatting — plain white cells with no borders or column styling
- Currency encoded inside cell number formats (not visible as a column)
- Inconsistent column ordering not suited for invoice registers

Business Impact

- Finance team spends 30–60 minutes per export on manual reformatting
- Risk of formula or copy-paste errors in manually reformatted files
- Inconsistent register formats across staff members
- Delayed reporting due to reformatting bottleneck
- No standardized output format for sharing with auditors or clients

Proposed Solution

Build a zero-installation, browser-based tool that accepts the Tally XLSX as-is and outputs a standardized, formatted Sales Register Excel file in one click.

4. Technology Stack Evaluation & Recommendation

Option 1: Browser-Based HTML Tool (Recommended)

Technology Stack: HTML5, CSS3, Vanilla JavaScript, SheetJS (XLSX parsing), ExcelJS (XLSX generation)

Advantages:

- Zero infrastructure — runs entirely in the browser, no server needed
- No installation required — works from a single HTML file
- 100% file privacy — data never leaves the user's device
- Instant deployment — share the HTML file or host on any static server
- No ongoing maintenance cost

Limitations:

- Cannot process very large files (100k+ rows) on low-spec devices
- No audit trail or central logging
- Customization requires editing the HTML file

Best Fit: Any organization needing fast, free, private file conversion with zero setup.

Option 2: Python Script

Technology Stack: Python 3, openpyxl / pandas

Advantages:

- Highly customizable — easy to add new columns or logic
- Can be scheduled or integrated into pipelines
- Handles large files efficiently

Limitations:

- Requires Python installation on each user's machine
- Non-technical users cannot operate without a GUI wrapper
- Harder to distribute and maintain across teams

Option 3: Server-Side Web Application

Technology Stack: Node.js / Python backend, cloud storage, web frontend

Advantages:

- Centralized — all users access one URL
- Audit logs and history possible
- Supports multi-user access control

Limitations:

- Requires server infrastructure and hosting cost
- Files uploaded to server — privacy risk for sensitive financial data
- Longer development and deployment time

Comparison Table

Criteria	Tally2Excel (Browser Tool)	Python Script	Server-Side Tool
Implementation Cost	Zero	Low	High
Infrastructure Required	None	Python Runtime	Server + Backend
User Technical Skill	None	Medium	Low (admin only)
Maintenance Effort	Very Low	Low	Medium
Offline Capability	Yes	Yes	No
File Privacy	100% Local	Local	Cloud Upload
Customization	Medium	High	High
Deployment Time	Instant	Hours	Days

Final Recommendation

Recommended Solution: Browser-Based HTML Tool (Option 1)

Justification:

- Addresses the core problem with zero cost and zero infrastructure
- Finance users can operate it without IT involvement
- Files never leave the local machine — ideal for confidential financial data
- The tool can be distributed as a single file via email, Google Drive, or intranet
- Provides a foundation to upgrade to a hosted web app if multi-user needs arise

5. System Architecture & Data Flow

Architecture Overview

The entire process runs client-side. No network requests are made during file processing.

Data Flow:

Tally XLSX → Browser File API → SheetJS Parser → Transform Layer → ExcelJS Generator → Download

Processing Steps

1. Input	User uploads Tally Sales Register XLSX via drag-and-drop or file picker
2. Parse	SheetJS reads file in-browser; extracts rows from row 12 onward (skipping 11 header rows)
3. Detect Currency	Cell number format (cell.z) is inspected to auto-detect INR, USD, EUR, GBP, etc.
4. Transform	Data is mapped to: Invoice No, Buyer Name, Invoice Date, Invoice Amount; rows are sorted by Invoice No ascending
5. Preview	First 5 rows are rendered in-browser as a preview table before download
6. Export	ExcelJS generates a styled XLSX with purple headers, alternating row fill, borders, frozen row, and auto-filter

Tally Column Mapping

The tool reads from the following fixed column positions in the Tally Sales Register export:

Tally Column	Array Index	Usage
Column A (Date)	index 0	Invoice Date — parsed from Excel serial or date string
Column B (Party Name)	index 1	Buyer Name — party ledger name from Tally
Column F (Voucher No.)	index 5	Invoice Number — voucher/reference number
Column G (Amount)	index 6	Invoice Amount — net amount with currency format

6. Output File Specification

Output Columns

Column Header	Source / Logic	Data Type
INVOICE NO.	Invoice number from Tally column F (col index 5)	Text / Number
BUYERS NAME	Party/buyer name from Tally column B (col index 1)	Text
INVOICE DATE	Transaction date formatted as D-MMM-YY (e.g. 5-Apr-25)	Date
INVOICE AMOUNT	Invoice value from Tally column G (col index 6); currency auto-detected	Number

Excel Formatting Applied

- Header row: Purple background (#6B2D8B), white bold text, center-aligned, row height 25
- Alternating row fill: Even rows — light purple (#F3EEF8); odd rows — white
- All cells: Single thin border (#CCCCCC) on all sides
- Header row frozen (row 1 stays visible while scrolling)
- Auto-filter enabled on all 4 columns (A1:D1)
- Date format: D-MMM-YY (e.g. 5-Apr-25)
- Amount format: Currency symbol preserved from source (e.g. '₹ 0.00', '\$ 0.00')
- Output filename: Sales_Register_Purple.xlsx

Currency Detection Logic

The tool inspects the number format code (cell.z) of the Amount column to auto-detect currency:

Condition in Format Code	Detected Currency	Excel Output Format
Contains INR or Rs	Indian Rupee	"₹ "0.00
Contains \$	US Dollar	"\$ "0.00
Contains € or EUR	Euro	"€ "0.00
Contains £ or GBP	British Pound	"£ "0.00
Contains 'Euro' (text)	Euro	"Euro "0.00

Condition in Format Code	Detected Currency	Excel Output Format
None matched	As-is from Tally	Original format preserved

7. How to Use the Tool

Pre-requisites

- Tally ERP / TallyPrime installed with Sales Register data
- Sales Register exported as XLSX format from Tally
- Any modern browser (Chrome, Edge, Firefox, Safari)
- The Tally2Excel HTML file (index.html)

Step-by-Step Usage

Step 1	Open index.html in any browser by double-clicking the file.
Step 2	Export the Sales Register from Tally as XLSX: Gateway of Tally → Display More Reports → Sales Register → Export → Excel (Spreadsheet).
Step 3	Drag and drop the exported XLSX file onto the drop zone, or click the drop zone to browse and select the file.
Step 4	Click the Convert File button. The tool will parse the file and show a 5-row preview table.
Step 5	Review the preview to confirm Invoice No, Buyer Name, Date, and Amount are correctly extracted.
Step 6	Click Download Purple Excel to save the formatted Sales_Register_Purple.xlsx to your device.

8. Key Features

Zero Backend / Offline	Entire processing happens in the browser. No internet required after loading. No files are uploaded anywhere.
Smart Header Skip	Tally exports contain 11 metadata rows before data. The tool automatically skips these and reads from row 12 onward.
Multi-Currency Support	Automatically detects INR, USD, EUR, and GBP from Tally's cell number format — no manual input required.
Auto Sort by Invoice No.	Output rows are sorted by Invoice Number in ascending order (numeric-first, then alphabetic fallback).
5-Row Preview	Before downloading, users can review a sample of converted records to confirm correctness.
Professional Excel Output	Output includes purple headers, alternating row shading, full cell borders, frozen header, and auto-filter — print-ready out of the box.
Drag & Drop Interface	Files can be dropped directly onto the interface for a fast workflow.
Single File Distribution	The entire tool is one self-contained index.html file. No npm install, no build step, no dependencies to manage.

9. Conclusion

The Tally2Excel Automation project delivers a lightweight, zero-infrastructure solution to a common pain point in Tally-based accounting workflows: the time-consuming manual reformatting of Sales Register exports.

By leveraging modern browser-based file processing (SheetJS for reading, ExcelJS for writing), the tool eliminates the need for any server, database, or installation — while producing a professionally styled Excel output that is immediately usable for reconciliation, client sharing, and audit purposes.

The solution is purpose-built for the current requirement and lays the foundation for future enhancements such as:

- Support for additional Tally report types (Purchase Register, Ledger Summary)
- Configurable column mapping for non-standard Tally exports
- Batch processing of multiple files
- Hosted web version with Google Drive integration

Key Outcomes Achieved

Outcome	Result
Manual reformatting effort	Eliminated (0 minutes vs 30-60 minutes)
Infrastructure cost	Zero
File privacy	100% local processing
User technical requirement	None — no installation needed
Output consistency	Standardized format every time
Deployment complexity	Single HTML file distribution